MIL-I-2703B
12 OCTOBER 1954

SUPERSEDING
MIL-I-2703
23 AUGUST 1951
MIL-I-2703A(SHIPS)
29 OCTOBER 1952
MIL-I-9009(USAF)
25 MAY 1953

MILITARY SPECIFICATION

INDICATOR, COMBUSTIBLE GAS, PORTABLE

This specification has been approved by the Department of Defense for use of the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope.—This specification covers indicators used for detecting and indicating the concentration of combustible gases and vapors.

1.2 Classification. — Portable combustible gas indicators shall be of the following types as specified (see 6.1):

Type A — For miscellaneous combustible gases or vapors.

Type B — For hydrogen only.

Type C — For paint and fuel material gases or vapors, remote head.

Type D — For specific combustible gases or vapors.

Type E — For miscellaneous combustible gases or vapors and atmospheres containing leaded gasoline vapors.

Type F — For miscellaneous gases or vapors and atmospheres containing leaded gasoline vapors; splashproof.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

NN-B-591 — Boxes, Fiberboard, Wood-Cleated (for Domestic Shipment).

NN-B-621 — Boxes, Wood, Nailed and Lock-Corner.

NN-P-515 — Plywood, Container Grade.

LLL-B-631 — Boxes; Fiber, Corrugated (for Domestic Shipment).

LLL-B-636 — Boxes, Fiber, Solid (for Domestic Shipment).

PPP-B-601 — Boxes, Wood, Cleated Plywood.

MILITARY

MIL-B-18 — Batteries, Dry.

JAN-P-100 — Packaging and Packing for Overseas Shipment: General Specification.

JAN-P-106 — Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.

JAN-P-108 — Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.

MIL-P-116 — Preservation; Methods of.

JAN-W-530 — Webbing, Cotton, Natural or in Colors.

MIL-I-983 —Interior Communication Equipment; General Specification for.

MIL-H-1142 — Hose, Rubber, Cutting and Welding (Fuel, Gases, Oxygen and Air).

315479—54—(M36)

Source: https://assist.dla.mil -- Downloaded: 20260-16T15:16Z
Check the source to verify that this is the current version before use.

MIL–L–3122 — Leather, Upper, Chrome, Retain (Grain Out) (Chrome-Vegetable Retanned).

MIL–E–5272 — Environmental Testing, Aeronautical and Associated Equipment; General Specification for.

MIL–L–10547 — Liners, Case, Waterproof.

MIL–I–15024 — Identification Plates, Information Plates, and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.

MIL–B–15071 —Book, Instruction, Preparation, Contents and Approval.

STANDARDS

MILITARY

MIL–STD–105— Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–108— Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment.

MIL–STD–129— Marking for Shipment and Storage.

MS15011 — Fittings, Hose, Welding and Cutting, Oxygen and Acetylene.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

BUREAU OF MINES

Bulletin 503—Limits of Flammability of Gases and Vapors.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Qualification. — The combustible gas indicators furnished under this specification shall be products which have been tested and have passed the qualification tests specified in section 4 (see 6.2).

3.2 General features.—The equipment shall be in accordance with the following paragraphs of Specification MIL–I–983 in addition to the requirements specified herein: (Whenever a requirement of Specification MIL–I–983 conflicts with a requirement of this specification, the requirement of this specification shall govern.)

Materials, general
Stressed or vital parts
Unstressed or nonvital parts
Wood
Magnesium
Aluminum
Brass and bronze
Iron and steel
Protection against corrosion (salt spray test is not required)
Gaskets
Plastics
Inflammable materials
Toxic compounds
Electrical parts, general
Resistors
Potentiometers
Fungus-inert materials
Restriction of materials
Electrical tapes
Enclosures, general
Accessibility
Degree of enclosure
Through bolting
Cover bolts
Threaded parts

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

Locking devices
Washers
Interchangeability and standardization
Shock, vibration and inclination (vibration only) (nonvital vibration test required)
Painting
Designation and marking
Design for production
Rounded corners and edges
Internal unit protection
Drilled and tapped holes
Welding
Soldering
Mounting of parts
Wiring
Workmanship and general inspection
General information

3.2.1 *Description.* — Each indicator shall include an indicating meter or meters, meter adjusting elements and switch, batteries and associated electrical circuitry all self-contained within a metal enclosure. A gas sampling system with accessories, if used or required for operation, shall be provided. The top surface of the enclosure shall include the indicating meter or meters, meter adjusting element knob or knobs or switch (or knob(s) and switch) all located so as to be convenient for operation. A sampling hose connection and an aspirator bulb connection, if used, shall be attached to the top or sides of the enclosure. The enclosure shall be furnished with a hinged top cover, made of the same material as the enclosure to provide protection to the indicator when it is being carried or stored. The top cover shall be so arranged that it cannot be closed unless the battery circuit of the indicator is open. The enclosure shall also be equipped with a shoulder type carrying strap or other arrangement which will permit the indicator to be hung about the neck, leaving the hands free to operate the controls. In the event a carrying case is provided in accordance with 3.2.7 the hinged top cover and carrying strap specified above need not be furnished.

3.2.2 *Operating principle and circuit.* — Indicators shall utilize a Wheatstone bridge type electrical circuit to detect the presence of combustible gases and vapors and shall employ a meter balancing resistance network with one or more heated filaments. The filaments shall be of platinum wire. In order to keep the explosion hazard to a minimum the energy of the power supply shall not exceed the energy obtainable from four number 6 dry cell batteries connected in series.

3.2.3 *Power supply.*

3.2.3.1 *Dry batteries.* — Unless otherwise specifically approved by the bureau or agency concerned, the power supply shall consist of dry batteries conforming to one of the types covered by Specification MIL–B–18. Batteries shall be of a type that when connected to the circuit shall not produce more energy than that specified in 3.2.2 but shall be of adequate capacity to withstand the life test specified in 4.3.8. Dry batteries shall be conveniently replaceable and this replacement shall permit the indicators to comply with the limits of accuracy specified in 3.3.4, 3.4.4, 3.5.4, 3.6.3, 3.7.3 and 3.8.4.

3.2.3.2 *Storage batteries.*—Storage batteries may be furnished for the power supply where approved by the bureau or agency concerned. Storage batteries shall be so designed that the electrolyte cannot spill and that the indicator shall meet the physical features and weight requirements specified in 3.2.4.

3.2.3.2.1 Suitable equipment for recharging the battery shall be furnished. Recharging equipment shall operate from 105-120 volt, 60 cycle, single phase, a. c. or 105-120 volt d. c. as specified (see 6.1).

3.2.4 *Physical features.* — The equipment shall be ruggedly constructed, all metal parts shall be of corrosion-resistant material or shall be treated or coated to resist corrosion, and operate satisfactorily after being subjected to the nonvital vibration test as specified in Specification MIL–I–983. After completion of the vibration tests the indicators shall operate freely and without error in excess of the accuracy specified herein.

Source: https://assist.dla.mil -- Downloaded: 202640-16T15:16Z
Check the source to verify that this is the currentversion before use.

3.2.4.1 *Temperature and humidity.* — The equipment shall be capable of satisfactory operation in any ambient temperature between the limits of +32° F. and 130° F. in any relative humidity up to 90 percent.

3.2.4.2 *Weight.*—The total weight of the indicator, including dry batteries but excluding the sampling hose and carrying case if used, shall not exceed 10 pounds. When a storage battery and charger is used the weight shall not exceed 15 pounds.

3.2.4.3 *Dimensions.* — The dimensions of the indicator, excluding the carrying case if supplied, shall not exceed the following:

Height	Width	Depth
Inches	Inches	Inches
8	7½	3½

3.2.4.4 *Inclination.*—The indicator, including the combustion chamber, the sensitive and comparison elements, and all internal components, shall be so arranged and designed that its accuracy of operation will not be adversely affected when the instrument is inclined up to 30° from the normal operating position.

3.2.4.5 *Carrying strap.* — The carrying strap may be fabricated of either cotton webbing conforming to Specification JAN–W–530, type III, Olive Drab No. 7, or chrome leather conforming to Specification MIL–L–3122. Either material shall be treated for mildew resistance.

3.2.5 *Safety.* — The combustion chamber or filaments of all types of indicators shall be provided with suitable flame arrestors in order to prevent flashbacks from igniting outside gases. The indicators shall pass the test specified in 4.3.9 without ignition of the gases.

3.2.6 *Spare filaments.* — Each indicator shall be supplied with one extra filament. Replacement of filaments shall not cause the indicator to exceed the limits of accuracy specified herein.

3.2.7 *Carrying case.*—In lieu of furnishing a hinged top cover (see 3.2.1) a separate carrying case (with cover) for the indicator complete with accessories may be provided. This case may be constructed of wood, metal, leather, plastic or any other suitable material and shall be properly treated or finished to resist deleterious effects of weather and moistures. The top cover shall be so arranged that it cannot be closed unless the battery circuit of the indicator is open. The carrying case strap shall be in accordance with 3.2.4.5.

3.2.8 *Operating instructions.*—Each indicator shall include a complete instruction pamphlet for the specific indicator concerned, which shall include operating and maintenance instructions, elementary or schematic wiring diagram of the indicator, and such pertinent information to assist the operator in the correct use and maintenance of the indicator. Included in the pamphlet shall be a chart showing a set of curves to interpret the meter indications for indicators designed for use with more than one type vapor or gas. An operating instruction plate shall also be provided on the top of the case in such a manner that it may be readily referred to while operating the indicator.

3.2.9 *Controls.*—The meter adjusting element knob(s) or switch (or both) for obtaining the initial zero setting of the indicating meter or meters shall be conveniently located to facilitate operation. These shall have sufficient tension or the knob shall be provided with a disengaging clutch in order to ensure that the meter setting cannot be accidentally disturbed. When a separate rheostat is used for the exclusive use of controlling voltage and the correct operating voltage is indicated on the indicating meter scale, a clutch type disengaging control knob need not be used.

3.2.10 *Zero setting and voltage adjustment.* —The electrical circuit of the complete indicator should be so arranged that the opera-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

tor, by reading the indicating meter, can adjust the voltage impressed upon the terminals of the Wheatstone bridge circuit to its proper operating point without disturbing the zeroing calibration. This will permit operating in a contaminated atmosphere without the necessity of purging the indicator with fresh air. This shall be accomplished in such a way that the operator may determine, without additional equipment, when the useful life of the batteries has been expended. The method of accomplishing this shall be included in the instruction pamphlet (see 2.3.8).

3.2.11 *Stability.*—The indicator shall meet the requirements of the stability test specified in 4.3.4. Suitable arrangements shall be provided to permit convenient restandardization or rebalancing utilizing uncontaminated air. The sampling hose if used may be disconnected from the indicator during the rebalancing operation.

3.2.12 *Aspirator and sampling hose (for use on all types of indicators except type C).*

3.2.12.1 The atmosphere under test shall be drawn through the combustion chamber by means of an aspirator bulb or small hand pump with restricted flow or check valves in order that the pumping action does not cause undue meter pointer oscillations.

3.2.12.2 The indicator enclosure shall be supplied with a suitable airtight fitting for attachment of the sampling hose. The hose connection shall be 9/16 inch—18 R. H. NF3 male threaded, the threaded portion, hole, and countersunk opening similar to that shown for piece 2 of Standard MS 15011 as specified in Specification MIL–H–1142. The countersunk opening shall be arranged to make an airtight coupling with the coupling end of the sampling hose connection.

3.2.12.3 The sampling hose, unless otherwise specified in the contract or order, shall consist of two lengths of 15 feet each of flexible hose, 3/16 inch inside diameter, the inside diameter of the hose conforming to Specification MIL–H–1142. The hose shall be flexible and made of nonabsorbent fabric reinforced rubber, rubber like, or synthetic rubber material impervious to petroleum products, alcohols and benzol. The ends of each length of hose shall be fitted with suitable airtight fittings similar to that shown on Standard MS 15011 as specified in Specification MIL–H–1142. One end of the hose shall be fitted with a hose connection similar to piece 5, holding captive, loosely, a connection nut similar to piece 4 to be used as a hose connection gland. The nut shall be threaded 9/16 inch—18 R. H. NF3. The other end of the hose shall also be fitted with a hose connection fitting similar to piece 4 except that it shall be arranged with a 9/16 inch—18 R. H. NF3 male threaded connection, the threaded portion, hole, and countersunk opening similar to that shown for piece 2. The countersunk opening shall be arranged to make an airtight coupling with the coupling end of a fitting similar to piece 5.

3.2.12.4 When specifically required by contract or order, a special length of hose shall be furnished with each indicator, either as additional or as a substitute for those furnished in accordance with 3.2.12.3. The hose shall be 5 feet long and arranged with fittings similar to those specified in 3.2.12.3.

3.2.12.5 *Probe tube.*—When specified in the contract or order (see 6.1), a probe tube of brass or aluminum tubing 3 feet long shall be furnished. One end of the probe tube shall be fitted with a female airtight fitting for attachment to the male threaded portion of the sampling hose or the fitting on the indicator enclosure. Fittings should be arranged in a similar manner to the arrangement used on the sampling hose by holding captive, loosely, a connection nut similar to piece 4 of Standard MS 15011, which should be used as a hose connection gland.

3.3 Type A indicator.

3.3.1 *General.*—Type A indicators shall be capable of safely, quickly, and accurately de-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

tecting all combustible gases and vapors associated with fuel oils or gasoline (not containing tetraethyl lead), alcohol and acetone vapors, illuminating or fuel gas, hydrogen and acetylene in mixtures with air or oxygen. The indicators shall be sensitive in detecting small quantities of the above explosive vapors or gases, and shall give a reliable indication to show when the mixture is in or above the explosive range or below the lower explosive limit.

3.3.2 *Scale.*—The scale range of the meter shall be adequate to register the deflections produced by the indicator for any air oxygen mixture of the gases or vapors specified in 3.3.1 in concentrations varying from zero to the lower limit of explosibility. The scale shall be marked in arbitrary units representing the composite range of zero to 100 percent of the lower limit of explosibility for these mixtures and shall contain not less than 20 uniform scale divisions. The portion of the scale above the 60 percent mark shall be colored red to denote the dangerous range and provide for a reasonable margin in the safety factor. The scale shall continue for a sufficient amount beyond the 100 percent mark to assure a significant reading when the mixture is beyond the lower explosive limit shown in table I.

TABLE I.—*Lower explosive limit for type A indicators.*

Gas and vapor	Lower explosive limit in air (percent by volume)
Hydrogen	4.0
Benzene	1.4
Acetone	2.55
Ethyl alcohol	3.3
Acetylene	2.3
Pentane	1.4

3.3.3 *Calibration.*—The calibration shall be such that the indication for any concentration, below the lower explosive limit of the gases and vapors specified in table I, shall not be lower than the true concentration, in terms of percent of lower explosive limit. A 60-percent of lower explosive limit concentration of any gas and vapor specified in table I shall not produce a reading on the scale lower than 60 percent or more than 100 percent of the lower explosive limit. The latter readings shall be within ±10 percent of the lower explosive limit of the gas and vapor under cons deration. The curve chart in the operating instructions pamphlet (see 3.2.8) should be referred to when the indicator is submitted for qualification approval to enable accurate determination of the concentrations, in percentage of the lower explosive limit, of the combustible gases specified in table I through an interpretation of the scale readings.

3.3.4 *Accuracy.*—The accuracy of the indicators shall be ascertained through interpretations obtained from the curve chart in the instruction pamphlet. The explosive condition, in percentage of the lower explosive limits, of mixtures being used to check the accuracy of the indicator shall be known from their identity and concentration. The scale reading produced by the known mixture shall be used as the index for interpreting the explosive condition, in percentage of the lower explosive limit, of the same type of combustible from the curve chart. The accuracy of the indicator shall be represented by the difference, in percentage of the lower explosive limit, between the known mixture and the value read from the curve chart, and this difference shall not exceed ±10 percent of the lower explosive limit.

3.3.5 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide when subjected to the tests specified in 4.3.6.

3.4 Type B indicator.—

3.4.1 *General.*—Type B indicators are for specific use in detecting the presence of explosive mixtures of hydrogen in air or oxygen. They shall indicate safely, quickly, and accurately the concentration of hydrogen in mixtures with air or oxygen below the lower explosive limit, and also shall indicate reliably the presence of concentrations within or above the explosive range.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

3.4.2 *Scale.*—The scale of the meter shall be graduated from 0 to at least 4 percent, reading directly the percentage of hydrogen in the atmosphere. The scale divisions shall not be larger than 0.25 percent. The portion of the scale representing from 60 percent of the lower explosive limit of hydrogen to the upper end of the scale shall be colored red and marked "Dangerous". The scale shall continue for a sufficient amount beyond the 4 percent mark to assure a significant reading when the mixture is beyond the lower explosive limit.

3.4.3 *Calibration.* — The calibration of a type B indicator shall be such that it will be capable of determining the amount of hydrogen present in less than explosive mixtures. The lower explosive limit of hydrogen in air shall be considered to be 4 percent by volume.

3.4.4 *Accuracy.*—The accuracy of a type B indicator shall be within 0.40 percent by volume of the actual amount of hydrogen present in the atmosphere being tested.

3.4.5 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide when subjected to the tests specified in 4.3.6.

3.5 Type C indicator.

3.5.1 *General.*—Type C indicators shall be capable of safely, quickly, and accurately detecting combustible gases or vapors associated with fuel oils, gasoline, or paints. The indicators shall be sensitive in detecting small quantities of methane and other hydrocarbon gases or vapors with the exception of acetylene, and also shall give a reliable indication to show when the mixture is in or above the explosive range or below the lower explosive limit shown in table II.

TABLE II.—*Lower explosive limit for type C indicators.*

Gas or vapor	Lower explosive limit in air (percent by volume)
Benzine	1.4
Acetone	2.55
Ethyl alcohol	3.3
Pentane	1.4

3.5.2 *Scale.*—The scale range of the meter shall be adequate to register the deflections produced by the indicator for any air oxygen mixture of the gases or vapors specified in 3.5.1 in concentrations varying from zero to the lower explosive limit. The scale shall be marked in arbitrary units representing the composite range of zero to 100 percent of the lower limit of explosibility for these mixtures and shall contain not less than 20 divisions. The portion of the scale above the 60 percent mark shall be colored red to denote "the dangerous" range and provide for a reasonable margin in the safety factor.

3.5.3 *Calibration.* — The calibration shall be such that the indication for any concentration below the lower explosive limit of the gases or vapors specified in table II shall not be lower than the true concentration, in term of percent of lower explosive limit. A 60-percent of lower explosive limit concentration of any gas or vapor specified in table II shall not produce a reading on the scale lower than 60 percent or more than 100 percent of the lower explosive limit. The latter readings shall be within ± 10 percent of the lower explosive limit of the gas or vapor under consideration. The curve chart in the operating instruction pamphlet (see 3.2.8) should be referred to when the indicator is submitted for qualification approval to enable accurate determination of the concentrations, in percentage of the lower explosive limit of the combustible gases specified in table II through an interpretation of the scale readings.

3.5.4 *Accuracy.*—The accuracy of the indicators shall be ascertained through interpretations obtained from the curve chart in the instruction pamphlet. The explosive condition, in percentage of the lower explosive limits, of mixtures being used to check the accuracy of the indicator shall be known from their identity and concentration. The scale readings produced by the known mixture shall be used as the index for interpreting the explosive condition in percentage of the lower explosive limit, of the same type of combustible from the curve chart. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

accuracy of the indicator shall be represented by the difference, in percentage of the lower explosive limit, between the known mixture and the value read from the curve chart, and their difference shall not exceed ± 10 percent of the lower explosive limit.

3.5.5 *Remote head.* — Type C combustible gas indicators shall be provided with a remote head such that gas or vapor mixtures may be sampled continuously at distances up to at least 30 feet from the indicator without pumping or other manipulation by the operator.

3.5.6 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide or in combustible gases and vapors containing tetraethyl lead when subjected to the tests specified in 4.3.6 and 4.3.7, respectively.

3.5.7 *Safety.*—In the event that the type C indicator is unsafe for use in any common combustible gas or vapor, such limitations shall be conspicuously marked on the identification plate or panel.

3.6 Type D indicator.

3.6.1 *General.*—Type D indicators shall be capable of safely, quickly, and accurately detecting and indicating the concentration of the combustible gases or vapors specified in the contract or order (see 6.1) in mixtures with air or oxygen. The indicators shall be sensitive in detecting small quantities of the specified gases or vapors and shall give a reliable indication to show when the mixture is in or above the explosive range or below the lower explosive limit.

3.6.2 *Scale.*— The scale of the meter shall be graduated to read directly the percentage of the specified gases or vapors in the atmosphere under test from zero to 100 percent of the lower explosive limit. The scale divisions shall not be larger than 0.25 percent. The portion of the scale representing from 60 percent of the lower explosive limit of the specified vapors or gases to the upper end of the scale shall be colored red and marked "Dangerous". The scale shall continue for a sufficient amount beyond the 100 percent mark to assure a significant reading when the mixture is beyond the lower explosive limit.

3.6.3 *Calibration and accuracy.* — Gas or vapors to be indicated and detected shall be as specified in the contract or order (see 6.1). The lower explosive limit in air (percent by volume) of the specific gas should also be as specified in the contract or order. Where no lower explosive limit of the vapor or gas is specified the limits given in Bureau of Mines Bulletin 503, Limits of Flammability of Gases and Vapors applies.

3.6.3.1 *Calibration.*—The calibration of a type D indicator shall be such that it will be capable of determining the amount of the specific gas or vapor present in less than explosive mixtures.

3.6.3.2 *Accuracy.*—The accuracy of a type D indicator shall be within ± 10 percent by volume of the actual amount of the specific gas or vapor present in the atmosphere being tested.

3.6.4 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide when subjected to the tests specified in 4.3.6. When the type D indicator is used for a specific gas or vapor containing tetraethyl lead the filaments shall not be affected when subjected to the tests specified in 4.3.7.

3.7 Type E indicator.

3.7.1 *General.*—Type E indicators shall be capable of safely, quickly and accurately detecting all combustible gases and vapors associated with fuel oils or gasoline (containing tetraethyl lead), alcohol and acetone vapors, illuminating or fuel gas, hydrogen and acetylene in mixtures with air and oxygen. The indicators shall be sensitive in detecting small quantities of the above explosive vapors or gases, and shall give a reliable indication to

Source: https://assist.dla.mil -- Downloaded: 202610-16T15:16Z
Check the source to verify that this is the current version before use.

show when the mixtures is in or above the explosive range or below the lower explosive limit.

3.7.2 *Scale.*—The scale range of the meter for a type E indicator shall conform to 3.3.2.

3.7.3 *Calibration and accuracy.*—The calibration and accuracy requirements for type E indicators shall be based on table I as concerns gas and vapors, with their corresponding lower explosive limits, as given in 3.3.3.

3.7.3.1 *Calibration.*—The calibration of a type E indicator shall conform to 3.3.3.

3.7.3.2 *Accuracy.*—The accuracy of a type E indicator shall conform to 3.3.4.

3.7.4 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide or in combustible gases and vapors containing tetraethyl lead when subjected to the tests specified in 4.3.6 and 4.3.7, respectively.

3.8 Type F indicator.

3.8.1 *General.*—Type F indicators shall be capable of safely, quickly, and accurately detecting all combustible gases and vapors associated with the liquids and gases shown in table III in mixtures with air or oxygen. The indicators shall be sensitive in detecting small quantities of the above explosive vapors or gases, and shall give a reliable indication to show when the mixture is in or above the explosive range or below the lower explosive limit.

3.8.2 *Scale.*—The scale range of the meter for a type F indicator shall be adequate to register the deflections produced by the indicator for any air oxygen mixture of the gases or vapors specified in table III. The dial shall consist of not less than 24 scale divisions reading from 0 to 120 percent of the lower explosive limit with figures at the 0, 20, 40, 60, 80, 100 and 120 percent graduations. The portion of the scale above the 60 percent mark shall be colored red to denote the dangerous range.

TABLE III.—*Lower explosive limit for type F indicators.*

Liquid gas and vapor	Lower explosive limit in air (percent by volume)
Jet and reciprocating engine fuels	
1. JP–1	
2. JP–3	
3. JP–4	
4. 100/130 octane aviation gasoline	
Ethyl alcohol	3.3
Hydrogen	4.0
Benzine	1.4
Methane	5.3
Leaded gasoline with octane rating of 80 and above	1.3
Hexane	1.25

3.8.3 *Calibration.* — The calibration of a type F indicator shall conform to 3.3.3 except that the gases and vapors specified in table III shall be used in lieu of those shown in table I.

3.8.4 *Accuracy.*—The accuracy of a type F indicator shall conform to 3.3.4 except that hexane shall be used as the known mixture as the calibrating medium and the scale reading so produced shall be used as the index for interpreting the explosive condition of the combustible under consideration from the curve chart.

3.8.5 *Filaments.*—The filaments shall not be affected by operation in hydrogen sulfide or in combustible gases or vapors containing tetraethyl lead when subjected to the tests specified in 4.3.6 and 4.3.7, respectively.

3.8.6 *Sampling hoses.*—Type F indicators shall be provided with two sampling hoses of 5- and 15-foot lengths in lieu of two 15 foot hoses as specified in 3.2.12.3. All other requirements specified in 3.2.12.3 shall apply.

3.8.7 *Identification marking.* — The complete indicator shall include an identification

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

plate with the complete information specified in Specification MIL–I–15024. Equipment assemblies, and parts shall be marked for identification in accordance with Specification MIL–I–983.

3.8.8 *Splashproof.*—The type F indicator shall be tested against the entrance of water in accordance with Standard MIL–STD–108.

4. QUALITY ASSURANCE PROVISIONS

4.1 General.—The sampling procedures for the various tests to be performed on the equipment shall be as follows: (The attention of the manufacturer is invited to the applicable paragraphs of Specification MIL–I–983 for information concerning these tests such as cost, where conducted, failures, test conditions, etc.)

4.1.1 *Qualification tests at a Government or other designated laboratory.*—Qualification tests shall be conducted at a laboratory designated by the Bureau of Ships. One indicator of each type covered in this specification shall be subjected to the following qualification tests. Tests shall be performed in the order listed.

Test	Requirement paragraph	Test paragraph
General inspection	3.3 to 3.8, inclusive	4.3.1
Temperature and humidity	3.2.4.1	4.3.2
Inclination	3.2.4.4	4.3.3
Vibration (nonvital)	3.2	MIL–I–983
Stability	3.2.11	4.3.4
Accuracy	3.3.4, 3.4.4, 3.5.4, 3.6.3, 3.7.3, 3.8.4	4.3.5
Hydrogen sulfide	3.3. to 3.8, inclusive	4.3.6
Battery and filament life	3.2.3	4.3.8
Tetraethyl lead (except types A and B)	3.5 to 3.8, inclusive	4.3.7
Safety	3.2.5	4.3.9
Calibration	3.3 to 3.8, inclusive	4.3.10
Splashproof (type F only)	3.8.8	———

4.1.2 *Preliminary tests.*—The first production unit of each type equipment covered in the contract or order shall be subjected to the following preliminary tests at a laboratory designated by the bureau or agency concerned. These tests will be required in the contract or order only where the successful bidders products have not passed the qualification tests listed in 4.1.1. These tests shall be performed, in general, in the order listed:

Test	Requirement paragraph	Test paragraph
General inspection	3.3 to 3.8, inclusive	4.3.1
Temperature and humidity	3.2.4.1	4.3.2
Inclination	3.2.4.4	4.3.3
Vibration	3.2	MIL–I–983
Stability	3.2.11	4.3.4
Accuracy	3.3.4, 3.4.4, 3.5.4, 3.6.3, 3.7.3, 3.8.4	4.3.5
Hydrogen sulfide	3.3. to 3.8, inclusive	4.3.6
Battery and filament life	3.2.3	4.3.8
Tetraethyl lead (except types A and B)	3.5 to 3.8, inclusive	4.3.7
Safety	3.2.5	4.3.9
Calibration	3.3 to 3.8, inclusive	4.3.10
Splashproof (type F only)	3.8.8	———

4.1.3 *Inspection tests at the place of manufacture.*—A sample number of equipments shall be selected in accordance with 4.1.3.1.2, 4.1.3.1.3, and 4.1.3.1.4, and each of the selected equipments shall be subjected to the following tests. Groups A and B tests shall be conducted at the place of manufacture. Group C test shall be conducted at a laboratory designated by the bureau or agency concerned. The tests shall be performed, in general, in the order listed and shall be conducted or witnessed by the Government inspector.

Test	Requirement paragraph	Test paragraph
Group A tests		
General inspection	3.3 to 3.8, inclusive	4.3.1
Stability	3.2.11	4.3.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

Test	Requirement paragraph	Test paragraph
Group B test		
Accuracy	3.3.4, 3.4.4, 3.5.4, 3.6.3, 3.7.3, 3.8.4	4.3.5
Group C tests		
Temperature and humidity	3.2.4.1	4.3.2
Inclination	3.2.4.4	4.3.3
Hydrogen sulfide	3.3 to 3.8, inclusive	4.3.6
Tetraethyl lead (except types A and B)	3.5 to 3.8, inclusive	4.3.7
Safety	3.2.5	4.3.9
Calibration	3.3 to 3.8, inclusive	4.3.10
Battery and filament life	3.2.3	4.3.8
Vibration	3.2	MIL–I–983
Splashproof (type F only)	3.8.8	---

4.1.3.1 *Sampling for inspection tests.*

4.1.3.1.1 *Inspection lot.*—All indicators of the same type presented for a Government acceptance inspection at one time shall be considered a lot. The lot may include the entire contract quantity or it may be the production of any convenient time period at the choice of the Government inspector. Prior to presenting the lot for inspection, the contractor shall have subjected each indicator to the group A and B tests and shall include in the lot presented for Government inspection only those indicators that have passed these tests.

TABLE IV.— *Sampling for group A tests AQL (approx.)=1.5 defective.*

Number of indicators in inspection lot	Number of indicators in sample	Number of indicators nonconforming on any group A test	
		Acceptance number	Rejection number
7 and under	All	--	--
8 to 15	7	0	1
16 to 40	10	0	1
41 to 110	15	0	1
111 to 300	25	1	2
301 to 500	35	1	2
501 and over	50	2	3

4.1.3.1.2 *Sampling for group A tests.* — Sample indicators shall be selected from each inspection lot by the Government inspector in accordance with table IV and shall be subjected to each of the group A tests listed in 4.1.3 with lot acceptance based on the sampling requirements of Standard MIL–STD–105.

4.1.3.1.3 *Sampling for group B tests.* — Sample indicators shall be selected from each inspection lot by the Government inspector in accordance with table V and shall be subjected to the group B test listed in 4.1.3 with lot acceptance based on sampling requirements of Standard MIL–STD–105.

TABLE V.—*Sampling for group B test.*

Number of indicators in inspection lot	Number of indicators in sample	Number of indicators nonconforming on any group B test	
		Acceptance number	Rejection number
3 and under	All	--	--
4 and 15	3	0	1
16 to 40	5	0	1
41 to 110	7	0	1
111 to 300	10	0	1
301 to 500	15	1	2
501 and over	25	2	3

4.1.3.1.4 *Sampling for group C tests.* — Group C tests will be required by the bureau or agency concerned when the basic design of the equipment or the material of a vital part has been changed. One complete equipment shall be selected by the Government inspector and subjected to each of the appropriate group C tests listed in 4.1.3 with acceptance based on 4.1.3.1.6.

4.1.3.1.5 *Acceptance, groups A and B tests.* —The results of each test shall be compared with specification requirements. Failure to conform to the specification requirements for any test shall be counted as a defect and the indicator shall be rejected. If the number of such nonconforming indicators in any sample exceeds the acceptance number for that sam-

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

ple, the lot represented by the sample shall be rejected. Rejected lots may be offered again for Government inspection tests provided all indicators in the lot have been retested by the contractor for the test(s) causing rejection and all noncompliance corrected.

4.1.3.1.6 *Acceptance, group C tests.* — The results of each test shall be compared with specification requirements. In the event of failure to conform to this specification for any group C test conducted, the contractor shall correct the cause of failure on future production units and repair the deficiency in all indicators produced on the contract or order.

4.2 Test conditions.

4.2.1 *Standard atmospheric conditions.* — Whenever the pressure and temperature existing at the time of test are not specified definitely it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg.) and at room temperature (approximately 77° F.). When tests are made with atmospheric pressure or room temperature differing materially from the above values, proper allowance shall be made for the diference from the specified condition.

4.2.2 *Tapping.*—Unless otherwise specified in the contract or order, the indicator shall be gently tapped or vibrated before a test reading is taken.

4.2.3 *Attitude.*—Unless otherwise specified in the contract or order the indicator shall be tested in normal operating position.

4.3 Test procedures.

4.3.1 *General.*—It shall be ascertained by careful examination that the indicators are in strict conformity with this specification in the matter of materials, workmanship, design, finish, dimensions, and weight.

4.3.2 *Temperature and humidity.*

4.3.2.1 *Temperature.*—The indicators shall be subjected to the following temperature cycle:

Hours[1]	Temperature
0 to 1	75° F. (room)
1 to 5	32° F.
5 to 11	131° F.
11 to 16	75° F.

[1] Time periods may be varied to suit laboratory test schedule.

4.3.2.2 *Humidity.*—The humidity test shall be conducted in accordance with the methods specified in Specification MIL–E–5272, procedure III.

4.3.3 *Inclination.*—The indicators shall be inclined 30° each side respectively of the normal operating position and remain within the accuracy requirements of the particular type of indicator under test.

4.3.4 *Stability.*—Each indicator under test shall be operated for 15 minutes on fresh air to allow the battery voltage to become relatively stable and then balanced on the zero point. The indicator then shall be operated for three consecutive 15-minute periods while burning a combustible gas-air mixture of known concentration and identity. At no time during any 15-minute period shall the variation in reading exceed 5 percent of the full scale value. The instrument shall be quickly and accurately rebalanced on fresh air before each 15-minute period.

4.3.5 *Accuracy.* — The accuracy of each indicator under test shall be within the limits specified herein for each type of indicator (see 3.3.4, 3.4.4, 3.5.4, 3.6.3, 3.7.3 and 3.8.4 as applicable).

4.3.6 *Hydrogen sulfide.* — Each indicator under test shall be operated for a period of 15 minutes in a hydrogen sulfide air mixture containing 2 percent by volume of hydrogen sulfide. The accuracy of the indicator shall not be impaired beyond the accuracy requirements of the indicator under test after burning the indicator filament as required above.

4.3.7 *Tetraethyl lead.* — Each indicator under test (except for types A and B) shall

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

be operated for a period of 15 minutes in an atmosphere representing less than half scale of gasoline vapors containing tetraethyl lead. The accuracy of the indicator shall not be impaired or the filament poisoned beyond the accuracy requirements of the indicator under test of burning the indicator filament as required above.

4.3.8 *Battery and filament life.* — Each indicator under test shall be operated for a total of 24 hours at approximately half scale reading on a mixture of illuminating gas and air, in 6 periods of 4 hours each. At the end of 24 hours service the indicator shall be serviceable and accurate within the limits specified herein subject to replacement of dry batteries. Batteries shall be serviceable for a continuous period of 2 hours without requiring replacement or recharging except for types A and B indicator, whose batteries shall have a serviceable life for a continuous period of 4 hours without requiring replacement or recharging.

4.3.9 *Safety.*—Each indicator under test shall be checked in oxyhydrogen and oxyacetylene mixtures of maximum explosibility; namely, mixtures of two parts hydrogen to one part oxygen and mixtures of one part acetylene to two and one-half parts oxygen.

4.3.10 *Calibration.*—Each indicator under test shall be checked for calibration before and after the completion of the tests specified in 4.3.6 to 4.3.9, inclusive. In addition, the effect of replacing the filaments shall be checked. The gas mixtures used for calibration shall be freed of water vapor.

4.3.10.1 *Types A, C, E, and F indicators.*—The calibration of types A, C, E, and F indicators shall be initially checked on such mixtures that the data obtained are suitable for comparison with the curve chart in the instruction pamphlet furnished with the indicator. Thereafter, several mixtures covering substantially the extremes relative to sensitivity of the indicator may be used as a basis for calibration checks in subsequent tests.

4.3.10.2 *Type B indicators.*—Type B indicators shall be checked on concentrations of hydrogen in air giving approximately ¼, ½, ¾, and full scale deflection.

4.3.10.3 *Type D indicators.*—Type D indicators shall be checked on mixtures of air with the combustible gases or vapors specified. The concentrations of the mixtures shall be such as to give approximately ¼, ½, ¾, and full scale deflection.

5. PREPARATION FOR DELIVERY

5.1 **Basic indicators and accessories.**—The general cleaning, preservation, packaging, and packing requirements specified herein shall be used to prepare the basic indicators and accessories for immediate use, storage, stock, or overseas shipment as specified (see 6.1).

5.1.1 *For immediate use.*—Indicators and accessories destined for shipment to domestic activities for immediate installation or use shall be prepared for delivery as follows:

5.1.1.1 *Cleaning, preservation and packaging.*—The cleaning, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.1.1.2 *Packing.* — Indicators and accessories shall be packed in wood cleated fiberboard, cleated plywood, nailed wood, corrugated or solid fiberboard boxes conforming to Specification NN–B–591, PPP–B–601, domestic type, NN–B–621, or the special requirements of Specification LLL–B–631 or LLL–B–636, respectively, at the option of the contractor provided that the various requirements of the container specification, such as maximum weight of the item packed and method of closure are complied with. The gross weight of wood boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed the weight limitations of the applicable box specification.

5.1.2 *For storage, stock, or overseas shipment.*—Indicators and accessories destined

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

for storage, stock, or overseas shipment shall be prepared for delivery as follows:

5.1.2.1 *Cleaning, preservation, and packaging.*—Indicator and accessories scheduled for storage, stock, or overseas shipment shall be cleaned by method C-1 and packaged by method IIb in accordance with Specification MIL-P-116. No contact preservative is required. Boxes used as containers for method IIb packages shall conform to the applicable specification listed in 5.1.1.2.

5.1.2.2 *Packing.*

5.1.2.2.1 *For domestic shipment, immediate use.*—Indicators, packaged as specified in 5.1.1, shall be packed in containers of the type, size, and kind commonly used for the purpose in a manner that will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Consolidated Freight Classification Rules, or regulations of other carriers as applicable to the mode of transportation.

5.1.2.2.2 *For stock or storage.*—Indicators and accessories destined for stock or for ultimate installation on a specific ship but temporarily stored shall be packed in accordance with 5.1.1.2.

5.1.2.2.3 *For overseas shipment.* — Indicators and accessories destined for direct overseas shipment shall be packed in wood cleated plywood, nailed wood, or fiberboard boxes or nailed wood crates conforming to Specification PPP-B-601, overseas type, Specification JAN-P-106, or symbol V3c or V3s of Specification JAN-P-108, respectively, at the option of the contractor, provided that the various requirements of the container specification, such as weight of item packed, are complied with. Plywood shall conform to type I or II, class 2 of Specification NN-P-515. Boxes shall be lined with a sealed water-proof case liner conforming to type 1, grade B, class 2 of Specification MIL-L-10547. Seams and closures shall conform to the appendix of Specification MIL-L-10547. Shipping containers shall be closed and strapped in accordance with the appendix of the applicable container specification. The gross weight of wood boxes shall not exceed 200 pounds. The gross weight of fiberboard boxes shall not exceed 70 pounds.

5.1.3 *Packing containers.*—When practicable the applicable containers, as specified in 5.1.1.2 and 5.1.2.2, shall be of uniform size and each shall contain an identical number of items. All containers shall be new and unused and shall be able to withstand storage, rehandling, and reshipment without the necessity for repacking. Cushioning methods and materials required to protect items from damage due to shock and vibration shall be in accordance with Specification JAN-P-100. Shredded newspaper or excelsior shall not be used.

5.2 Publications.—Two copies of the instruction pamphlet shall be packed within the shipping container holding the indicator. Bulk quantities of instruction pamphlets shall be packed in accordance with Specification MIL-B-15071.

5.3 Marking.—In addition to the requirements specified in Standard MIL-STD-129, each unit or interior package and each shipment shall be clearly and legibly marked with the electrical and operational characteristics (as applicable).

5.4 Pilot packs.—When specified (see 6.1) pilot packs shall be prepared and subjected to the applicable tests specified in Specification MIL-P-116.

6. NOTES

6.1 Ordering data. — Procurement documents shall specify the following:

(a) Title, number, and date of this specification.

(b) Type of equipment required (see 1.2).

(c) Quantity of items required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

(d) Power supply required for recharging storage batteries, if used (see 3.2.3.2.1).

(e) The specific combustible gases or vapors for which type D indicators are intended (see 3.6).

(f) Probe tube if required (for types A, D, E, and F only) (see 3.2.12.5).

(g) Quantity of instruction books.

(h) Whether or not preliminary tests are required (see 4.3.1).

(i) Whether equipment is to be prepared for immediate use, storage, stock or overseas shipment (see 5.1).

(j) When pilot pack is required (see 5.4).

6.2 In the procurement of products requiring qualification, the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on a Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained from the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C.

6.3 Nomenclature. — The name of parts shall conform to "Federal Item Identification Guides for Supply Cataloging", Handbook H 6–1, published by the Standardization Division (Supply and Logistics), Office of the Assistant Secretary of Defense.

6.4 The packaging, packing, and marking specified herein apply only to direct shipment to the Government and are not intended to apply to contract or orders between the manufacturer and the prime contractor.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Custodians:
Army—Chemical Corps
Navy—Bureau of Ships

Other interest:
Army—EOT
Navy—OrY.

Source: https://assist.dla.mil -- Downloaded: 202640-16T15:16Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers, vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

☆ U.S. GOVERNMENT PRINTING OFFICE 1973—714-866/3983 2-1

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID



OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
Edgewood Arsenal
ATTN: SAREA-TS-S
Edgewood Arsenal, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T15:16Z
Check the source to verify that this is the current version before use.